

CLI reference — ‘qf-agent’

QF-AgentOS — An honest benchmark and evidence harness for quantum finance

QF-AgentOS contributors

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Eleven commands. Every command is deterministic (same spec + seed $\Rightarrow$ same result) and prints the disclaimer; nothing places trades or moves money.

```
qf-agent --help
qf-agent <command> --help
```

Command	Autonomy	Purpose
<code>solve</code>	L2 (L3 with <code>--approve</code>)	Run the full pipeline $\rightarrow$ decision + evidence bundle
<code>explain</code>	L0	Understand + formulate only. No solving
<code>plan</code>	L1	Experiment plan (reduction, backend, cost). No execution
<code>arena</code>	L2	Benchmark every family $\rightarrow$ honest leaderboard
<code>estimate</code>	L2	Quantum amplitude estimation vs classical MC
<code>simulability</code>	L2	Tensor-network classical-simulability check
<code>backends</code>	—	List backends + real availability
<code>skills</code>	—	List installed Quantum Skills
<code>runs</code>	—	List recorded runs from the evidence store
<code>serve</code>	—	Run the REST API + Studio
<code>version</code>	—	Version

`solve`

Runs all 9 agents and writes an evidence bundle.

```
qf-agent solve examples/collateral-allocation.yaml
qf-agent solve spec.yaml --out evidence/ --approve --quiet
```

Option	Default	Meaning
--out, -o	evidence	Evidence output directory
--yes, --approve	off	Approve paid/irreversible steps (required for real QPU)
--quiet, -q	off	Suppress the per-step trace

Exit codes: 0 success · 2 bad spec · 3 missing extra · 4 problem infeasible · 5 backend error.

Prints the FINAL DECISION block and writes manifest.json, report.md, model_card.md. A failed quantum step is recorded and the run **continues** on the classical baseline — it never silently aborts.

explain

L0 — validate the spec, enumerate candidate formulations (LP / MILP / QUBO / Ising) and what each can and cannot represent. No solving.

```
qf-agent explain examples/collateral-allocation.yaml
```

plan

L1 — the experiment plan: reduced instance size, QUBO density, backend target (or abstention + reasons), estimated depth/cost, and the **encoding losses the verifier will re-check**. No execution.

```
qf-agent plan examples/settlement-netting.yaml
```

arena

Benchmarks every bundled example across backends and prints the honest leaderboard. This is the headline artifact — see [FINDINGS.md](#).

```
qf-agent arena
qf-agent arena --out arena/      # also writes arena.md + arena.json
```

7 problems · 0 quantum advantage · 1 parity · 6 classical preferred.

estimate

Quantum Amplitude Estimation (MLAE, real Grover operator) vs exact summation and classical Monte Carlo, on a discretised loss distribution — plus an honest resource analysis. Needs the qiskit extra.

```
qf-agent estimate --qubits 4          # expected loss
qf-agent estimate --qubits 4 --tail 0.7 # VaR-style P(loss > 0.7)
qf-agent estimate --qubits 5 --out out/ # writes estimate.json
```

Option	Default	Meaning
--qubits, -m	4	Distribution qubits (2^m loss levels)
--tail	—	Tail threshold → exceedance probability; omit for expected loss
--shots	200	Shots per Grover power
--seed	7	Deterministic
--out, -o	—	Write estimate.json

simulability

Tensor-network baseline: is this problem's QAOA circuit classically simulable by an MPS? Reports entanglement entropy, the bond dimension for a target fidelity, truncated-MPS fidelity, and the honest verdict. Needs qiskit; optimisation families only.

```
qf-agent simulability examples/collateral-allocation.yaml
qf-agent simulability spec.yaml --reps 2 --fidelity 0.999
```

backends

Every backend and whether it is available **right now** (truthful discovery — a credentialed backend reports exactly what is missing).

```
qf-agent backends
```

```
qaoa_sim          yes    qiskit statevector QAOA
qaoa_ibm          no     set QF_IBM_TOKEN (and optionally QF_IBM_INSTANCE/QF_IBM_BACKEND)
dwave_hybrid      no     install qf-agentos[dwave] (dwave-ocean-sdk)
```

skills

Installed Quantum Skills (a skill = a skill.yaml manifest declaring a problem family, its methods, and its verification checks).

```
qf-agent skills
qf-agent skills --dir ./my-skills
```

runs

List recorded runs from the file evidence store.

```
qf-agent runs
qf-agent runs --out evidence/
```

serve

Run the REST API and the bundled Studio. Needs the server extra. See [API.md](#).

```
qf-agent serve --host 0.0.0.0 --port 8000
# equivalently: uvicorn qf_agentos.api:app
```

version

```
qf-agent version
```

Running on real hardware

```
export QF_IBM_TOKEN='...' # your shell only; never committed
qf-agent backends        # confirm qaoa_ibm available=yes
qf-agent solve spec.yaml --approve # spec needs qpu_backend + autonomy L3
```

Real QPU execution requires **all** of: credentials, qpu_backend: ibm|dwave, autonomy_level: L3, sufficient max_qpu_budget_usd, and --approve. Missing any one falls back to the simulator with a recorded reason. Full runbook: [real-qpu.md](#).